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New issue



ND Pseudo Waypoints #856

Closed

#7080

Ares1378 opened on Sep 25, 2020

Last edited by Ares1378 ▾ ⋮

Mod Version

Master

Describe the bug

Pretty much all of these items are missing on the ND besides the TOC and TOD items.

To Reproduce

1. go for a flight from point a to point B

2.

3.

Expected behavior

The following items should be displayed on the ND for the following reasons:

These predictions consist of symbols positioned along the lateral flight plan (NAV mode engaged) or the track line (NAV mode not engaged). These symbols (named as pseudo waypoints) and their meanings are:

Pseudo waypoint	Definition
	Level symbol at the position (top of climb or level-off) where the aircraft will reach: - The FCU selected altitude (blue) or - The constrained altitude, if it is more restrictive than the FCU altitude and if appropriate modes are engaged (magenta).
	Top of descent or continue descent symbol: - Top of descent (always white) - Continue descent symbol (white if DES is not armed, blue if it is).
	Start of CLIMB symbol: - White if CLB is not armed - Blue if CLB is armed.
	Intercept point symbol: The point where the aircraft is predicted to intercept the descent path, if there is any vertical deviation when the aircraft is in DES mode (white if DES is not engaged, blue if it is).
•	Speed change symbol: The point at which the aircraft will initiate an automatic ACCEL or DECEL from current speed to a new computed speed if it encounters a SPD LIM, SPD CSTR, or HOLDING SPD (magenta).
	Decelerate point symbol: - Indicates the point at which the aircraft is predicted to decelerate for approach (and thus switch to the approach phase) - Magenta, if in managed speed and NAV or approach mode is engaged - White, if in selected speed or HDG/TRK mode - Automatic deceleration only occurs when displayed in magenta.

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Pseudo waypoint	Definition
	ALT CSTR symbol set around the constrained waypoint: - Magenta, when the ALT CSTR is predicted to be met - Amber, when the ALT CSTR is predicted to be missed - White, when the ALT CSTR is not taken into account by the FMGS, and the NAV mode is engaged.
 (10 45)  (ETP)	Time marker and equitime point symbols appear in green to indicate where the aircraft reaches the time marker or equitime point.
	Energy circle symbol (green arc) centered on the aircraft position and oriented to the current track line. Represents the Required Distance to Land. Only displayed if the lateral guidance mode is heading or track, and the current FMS flight phase is in cruise, descent or approach, and the aircraft is within 180 NM of the destination.
Crosstrack error XX.XR or XX.XL (X is a number)	The crosstrack error displays the lateral deviation between the aircraft position and the track of the F-PLN active leg. The value is limited to 99.9 NM left or right.
INTCPT	Intercept waypoint is displayed on the ND at the point at which the present track intercepts the F-PLN.

The predicted time of arrival at the TO WPT is located in the upper right-hand corner of the ND. It assumes direct distance from the aircraft position to the TO WPT and assumes current ground speed will be constant.



As a general rule, the ND indicates what the aircraft will fly, with the current active FG modes.

For example:

- The continuous green line on the ND represents the track the aircraft is currently flying:
 - If HDG/TRK is engaged, the track line is green and the flight plan is dashed
 - If NAV mode is engaged, the green line is the flight plan.
- If the speed target is manually selected, the speed-change symbol is no longer displayed because it will not be taken into account.
- When the aircraft is not following the vertical flight plan (OP CLB, OP DES, V/S) but the NAV mode is engaged, the system disregards any altitude constraints and puts white circles around the waypoints that have these constraints and positions level symbols accordingly.
- Pseudo waypoints are adjusted each time predictions are updated.

Better look at the symbols BUT PLEASE REFER TO THE DOCUMENTS ABOVE FOR DESCRIPTION:

SYMBOL	NAME	MODE	DEFINITION
	Waypoint (W, G, M)	NAV ARC PLAN	The TO waypoint displays white. All other flight plan waypoints display green. Waypoints not in the flight plan display magenta when WPT button is selected on.
	Start of Climb (B, W)	NAV ARC PLAN	The point on the flight plan where the airplane is predicted to start the climb. Blue when CLB armed; white when not armed.
	Level Off Point (B, M)	NAV ARC PLAN	When blue, the point where the airplane will reach the FCU-selected altitude. When magenta, the point where the airplane will reach a VNAV altitude constraint if not set in the FCU.
	Top of Descent (B, W)	NAV ARC PLAN	The top of descent or continue to descend symbol. Blue when DES/FINAL armed; white when DES/FINAL not armed.
	Intercept Point (B, M)	NAV ARC PLAN	Indicates the point where the airplane meets the FMGC-computed vertical profile. If the intercept is from above the vertical path, the descent is based upon using 1/2 speed brakes. Blue when DES is engaged; white when only NAV is engaged.
	Speed Change (M)	NAV ARC PLAN	Indicates the point where the airplane will initiate an automatic acceleration or deceleration from current speed to new computed speed for SPD LIM, SPD CSTR or HOLDING SPEED.

SYMBOL	NAME	MODE	DEFINITION
	Decelerate Point (M)	NAV/ARC PLAN	Indicates where the airplane will start an automatic deceleration toward VAPP (switches into the APP phase).
	Altitude Constraint (M, A, W)	NAV ARC PLAN	Indicates an altitude constraint. Magenta if constraint is predicted to be met, amber if predicted to be missed, and white if guidance is not taken into account and NAV mode is engaged.
	Energy Circle (G)	NAV ARC PLAN	Indicates the radius corresponding to the required distance to land from present position. Only displayed in descent and approach phase when in HDG or TRK modes.
FLIGHT PLAN LINES			
	Active Flight Plan Route (G)	NAV ARC PLAN	Displays the primary flight plan actually being followed when NAV mode is engaged. Dashed when the NAV mode is not engaged.
	Missed Approach (B,M)	NAV ARC PLAN	Displays when a missed approach waypoint is displayed on the onside MCDU and the go-around phase is not active.
	Secondary Flight Plan (W)	NAV ARC PLAN	Displays the secondary flight plan when any SEC F-PLN page is displayed on the onside MCDU. Active flight plan remains displayed.

FLIGHT PLAN LINES			
SYMBOL	NAME	MODE	DEFINITION
	Temporary Flight Plan (Y)	NAV/ARC PLAN	Displays TMPY flight plan prior to INSERT.
	Engine Out Flight Plan (G, Y)	NAV ARC PLAN	Displays in green when EO flight plan is active; yellow when EO flight plan is not active.
PATH SYMBOLS			
	Procedure Turn or Holding Pattern (W)	NAV ARC PLAN	Displayed on the 160 NM and 320 NM scales or when the turn is beyond the next leg.
	Holding Pattern (W)	NAV/ARC PLAN	Displayed when range is 80 NM or less and the turn is in the active or next leg.
	Runway (W)	NAV/ARC PLAN	Displays selected runway as oriented runway symbol. Drawn to scale (paved length) if 10, 20, or 40 NM range is selected.
NAVAIDS			
	DME or TACAN (G,W, B, M)	NAV ARC PLAN	Displays green if a current waypoint, white if the TO waypoint, blue when tuned for display (automatically or manually), and magenta when not part of the flight plan but displayed when the VOR.D and/or NDB buttons are selected on.

NAVAIDS			
SYMBOL	NAME	MODE	DEFINITION
	VOR (G,W, B, M)	NAV ARC PLAN	Displays green if a current waypoint, white if the TO waypoint, blue when tuned for display (automatically or manually), and magenta when not part of the flight plan but displayed when the VOR.D and/or NDB buttons are selected on.
	VOR / DME (G,W, B, M)	NAV ARC PLAN	Displays green if a current waypoint, white if the TO waypoint, blue when tuned for display (automatically or manually), and magenta when not part of the flight plan but displayed when the VOR.D and/or NDB buttons are selected on.
	NDB (G,W, B, M)	NAV ARC PLAN	Displays green if a current waypoint, white if the TO waypoint, blue when tuned for display (automatically or manually), and magenta when not part of the flight plan but displayed when the VOR.D and/or NDB buttons are selected on.
MISCELLANEOUS			
	Airport (M, W)	NAV ARC PLAN	Displays white when the runway is not specified. Displays magenta for airports that are not part of the flight plan (regular, refueling, alternate, and emergency) when ARPT button is selected on.

MISCELLANEOUS			
SYMBOL	NAME	MODE	DEFINITION
	Cross Track Error (G)	NAV ARC PLAN	Displays left or right lateral deviation from the active leg of the flight plan in NM. Not present when PPOS is the FROM waypoint.
	Weather Radar Returns	ILS/VOR NAV ARC	Areas of the highest intensity display in red, lesser intensity in amber, and lowest intensity green. Turbulence is displayed in magenta.
	Predictive Windshear (Black, R, A)	ILS/VOR NAV ARC	Indicates windshear area as a black and red icon and two amber radial lines.
	EGPWS (R, Y, G, Black, M)	ILS VOR NAV ARC	Indicates threatening terrain, with red being the most threatening, followed by yellow then green. Black indicates terrain is 2000 feet below airplane and/or within 400 feet of the runway elevation of the nearest airport. Magenta indicates terrain data is unavailable.
TCAS			
	TCAS RA Traffic (R)	ILS VOR NAV ARC	With TA/RA or TFC selected, displays RA intruder within 15 to 35 seconds of closest point of approach (CPA). Inhibited below 1100 feet AGL on climb and 900 feet on descent. Range must be 40 NM or less on some airplanes.

TCAS			
SYMBOL	NAME	MODE	DEFINITION
	TCAS TA Traffic (A)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, displays TA intruder within 20 to 48 seconds of CPA. TA traffic only displayed by non-altitude reporting transponders even if intruder meets RA criteria. Range must be 40 NM or less on some airplanes. Inhibited below 500 feet AGL.
	TCAS Proximate Traffic (W)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, displays nonthreatening intruder within 1200 feet vertically and 6 NM laterally. Symbol alone indicates traffic without altitude reporting and must be assumed to be co-altitude. This symbol is visible only if a TA or RA is active and the range selected is 40 NM or less on some airplanes.
	TCAS Other Traffic (W)	ILS VOR NAV ARC	With TFC selected, displays nonthreatening traffic within about 30 NM and 2700 feet vertically. Symbol alone indicates traffic without altitude reporting and must be assumed to be co-altitude. Range must be 40 NM or less on some airplanes.
	TCAS Vertical Speed Arrows (R, A, W)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, indicates traffic is climbing ↑ or descending ↓ at least 500 FPM. Arrows not displayed indicates vertical motion less than 500 FPM. Range must be 40 NM or less on some airplanes.

TCAS			
SYMBOL	NAME	MODE	DEFINITION
	TCAS TA Traffic (A)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, displays TA intruder within 20 to 48 seconds of CPA. TA traffic only displayed by non-altitude reporting transponders even if intruder meets RA criteria. Range must be 40 NM or less on some airplanes. Inhibited below 500 feet AGL.
	TCAS Proximate Traffic (W)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, displays nonthreatening intruder within 1200 feet vertically and 6 NM laterally. Symbol alone indicates traffic without altitude reporting and must be assumed to be co-altitude. This symbol is visible only if a TA or RA is active and the range selected is 40 NM or less on some airplanes.
	TCAS Other Traffic (W)	ILS VOR NAV ARC	With TFC selected, displays nonthreatening traffic within about 30 NM and 2700 feet vertically. Symbol alone indicates traffic without altitude reporting and must be assumed to be co-altitude. Range must be 40 NM or less on some airplanes.
	TCAS Vertical Speed Arrows (R, A, W)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, indicates traffic is climbing ↑ or descending ↓ at least 500 FPM. Arrows not displayed indicates vertical motion less than 500 FPM. Range must be 40 NM or less on some airplanes.

TCAS			
SYMBOL	NAME	MODE	DEFINITION
10.3NM +01↑ 12.4NM -02 ↓	TCAS No Bearing Data (A, R)	ILS VOR NAV ARC	Up to two displayed when no bearing information is available. Displays distance in NM, altitude, and vertical direction at the bottom of the ND. Displays in amber or red according to threat level.
10.3 NM	TCAS No Bearing & No Alt Data (A, R)	ILS VOR NAV ARC	Displays distance in NM at the bottom of the ND when no bearing or altitude information is available. Displays in amber or red according to threat level.
TCAS FAIL	TCAS Failure Indication (R)	ILS VOR NAV ARC	Indicates TCAS failure. Flashes for 9 seconds, then remains steady. NAV TCAS FAULT appears on upper ECAM.
TCAS: OFF	TCAS Off Indication (B)	ILS/VOR NAV ARC	Indicates STBY has been selected or both transponders have failed.
TA ONLY	TCAS TA Only Indication (A, W)	ILS VOR NAV ARC	Indicates TA only mode is selected or the RA function is inhibited or failed. With TA/RA selected, the message appears automatically during climb until 1100 feet AGL and on descent below 900 feet AGL when the TCAS automatically switches to the TA ONLY mode.

TCAS			
SYMBOL	NAME	MODE	DEFINITION
TCAS Offscale	TCAS Offscale (A for TA) (R for RA)	ILS VOR NAV ARC	With TFC selected, RA or TA traffic is outside the traffic area covered by the selected ND range.
+05 -05 00	TCAS Relative Traffic (W)	ILS VOR NAV ARC	With TA, TA/RA, or TFC selected, displays relative traffic altitude in hundreds of feet. For an intruder above the airplane, the relative altitude will be placed over the traffic symbol preceded by an + sign. For intruders co-altitude or lower, the relative altitude will be placed below the target symbol preceded by - sign (except 00, which had no sign). Range must be 40 NM or less on some airplanes.

BELOW ARE THE CORRECT MCDU PERF PAGE PHASE CHANGES:

The vertical flight plan is divided into flight phases. For each phase, the FMGS computes the optimum speed or Mach Profile. These flight phases are:

Preflight - Takeoff - Climb - Cruise - Descent - Approach - Go-Around - Done.

FLIGHT PHASES	OPTIMUM SPEED PROFILE	SWITCHING CONDITIONS TO NEXT PHASE
PREFLIGHT	/	SRS takeoff mode engaged and $N1 > 85\%$ ($EPR \geq 1.25$) or Ground Speed > 90 kt
TAKEOFF	$V2$ ($V2 + 10$)	At acceleration altitude or by engagement of another vertical mode.
CLIMB	ECON CLB SPD / MACH	Reaching cruise FL
CRUISE	ECON CRZ MACH	No step descent, and distance to destination < 200 NM, or all engines operative and selected altitude below Max [FL 200, highest DES ALT CSTR]

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FLIGHT PHASES	OPTIMUM SPEED PROFILE	SWITCHING CONDITIONS TO NEXT PHASE
DESCENT	ECON DES MACH / SPD	- Overflying (DECEL) pseudo waypoint with NAV (or LOC*/LOC) mode engaged and altitude $< 9\,500$ ft AGL - Manual activation of the approach phase.
APPROACH	VAPP (GS Min)	1. To Go-Around: When thrust levers at TOGA detent, or 2. To Done: 30 s after landing, or 3. To Climb: When inserting a new CRZ FL.
GO-AROUND	VAPP or current SPD, whichever is greater. Green Dot at ACC ALT	1. To Approach: Manual activation of the approach phase, or 2. To Climb: Above acceleration altitude, modification of the destination airport by: - Selection of the ALTN, or - Insertion of a NEW DEST, or - Insertion of a SEC F-PLN with a destination airport different from the destination airport of the active F-PLN.
DONE	/	To preflight: When INIT or PERF key depressed.

Note: During the preflight phase, the flight crew inserts the flight plan, which includes all data needed for the flight.

During the Done phase, the FMGC erases the data entered for the flight. If the descent or the approach phase is inadvertently activated (manual approach phase activation, for example), the flight crew may reselect a CRZ FL on the PROG page to reactivate the CRZ phase.

Actual behavior

References

Additional context

Was this working before/when did the issue start occurring?

Is this a problem in the vanilla unmodded game?

Discord username (if different from GitHub):



-  Ares1378 added Bug on Sep 25, 2020
-  lousybyte added ND on Sep 28, 2020
-  derl30n changed the title Missing/wrong indication the ND in regards to flight plan and waypoints and I have added the correct Phase change sequence of the MCDU[BUG] [REQUEST] ND Pseudo Waypoint Tracker on Dec 31, 2020
-  derl30n changed the title [REQUEST] ND Pseudo Waypoint Tracker [REQUEST] ND Pseudo Waypoints on Dec 31, 2020
-  derl30n added Request and removed Bug on Dec 31, 2020
-  derl30n mentioned this on Jan 8, 2021
-  [REQUEST] Add energy circle #2930



tellurics on Feb 6, 2021

Any updates so far?





Benjozork on Feb 6, 2021

Member

This is gonna be part of VNAV and the reworked AP

 1

-  Benjozork mentioned this on May 20, 2021
-  AP TOD implimation #4860



Swaluver88 on May 20, 2021

looking really forward to this, it will put this as the first Aircraft in MSFS with a realistic AP CLB and Descent profile system



-  tracernz mentioned this on Jun 2, 2021
-  Altitude Range Arc (ARC) - AKA Green Banana not working #4958

-  derl30n added FPM FMS - Predictions on Oct 14, 2021

-  tracernz mentioned this on Feb 17, 2022
-  Descent/climb target #6734

-  derl30n changed the title [REQUEST] ND Pseudo Waypoints ND Pseudo Waypoints on Apr 10, 2022

-  derl30n mentioned this on Apr 10, 2022
-  T/C and T/D pseudo-waypoints #307

-  tracernz mentioned this on Jul 28, 2022
-  Target Altitude indidactors #7372

-  BlueberryKing mentioned this on Mar 29, 2023
-  feat: vertical navigation v1 #7080

-  BlueberryKing closed this as completed in #7080 on May 11, 2023



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